SUMMARY OF UPDATES

AUTONOMOUS SG 2026 | PROBLEM STATEMENT UPDATE | 31 OCT 2026, 12:45 PM

Read this first

- The challenges have **not** changed. The update answers open questions, fills data gaps and adds new files.
- **Where the update differs from the original, follow the update.** The judging rubrics are unchanged.
- Same clarifications everywhere: this document, the updated problem statement PDFs, the event page and Discord.
- Full updated statements: Airbus PDF (updated) · NUS Hackers PDF (updated) · event page · datasets Drive
- **All Airbus challenges:** judges accept reasonable assumptions you state clearly in your demo. Public external data is allowed if cited and fetched by a script. Time discipline is scored on your **first valid submission**.

At a glance

Challenge	Issue	Update	New file
Airbus 1: Satellite collisions	Units unclear; tier readable from risk column; test set has no labels	Metres confirmed; predict final tier from CDMs ≥ 2 days out; evaluate on train split	Event labels (xlsx)
Airbus 2: Air traffic	Flight delay file missing; no gate, runway, fuel or crew data; “turn” undefined	flights.csv added; constraints workbook; turn defined	flights.csv, constraints (xlsx)
Airbus 3: SAF supply chain	Capacity, yield, demand, OpEx and time basis undefined; airport names ambiguous	All parameters fixed; 4 airports with IATA and coordinates	Parameters (xlsx), airports (csv)
Airbus 4: Regulated info	431 not 432 rows; proposals not ADs; no revision data; ~150 pages not 1,000+	Revision chains; full text allowed from Federal Register	AD index (xlsx)
Airbus 5: Weather & delay	No wind direction; sample can’t give hourly AAR; leakage risk	Hourly wind added; AAR optional; 3-hour lag rule	Hourly weather (csv), guide (xlsx)
NUS Hackers	No legality, build-time or sourcing rules	Submission checklist and rules added	—

Airbus Challenge #1: Satellite collision avoidance

- **Units:** **miss_distance** and **relative_position_*** are in metres. Thresholds are 1,000 m and 5,000 m.
- **Task:** predict each event's **final** tier using only CDMs with **time_to_tca** ≥ 2 days.
- **Evaluation:** test_data has no outcomes. Report metrics on a held-out split of train_data by event_id; use test_data for demo predictions.
- **event_id:** restarts in each file. Never join train and test on it.
- **risk = -30:** placeholder for near-zero probability (about 41% of rows).
- **Dashboard:** no orbits or clock times in the data; plot against time_to_tca instead.
- **New:** challenge1_event_labels.xlsx: final risk, miss distance and tier for all 13,154 train events.

Airbus Challenge #2: Air traffic decisions

- **Missing data:** flights.csv added: 358,774 rows from 2015 with **WEATHER_DELAY** > 15 or **LATE_AIRCRAFT_DELAY** > 20.
- **Turn:** each of the 3 agents sends one structured message. Consensus within 3 turns, otherwise the deterministic fallback decides.
- **Operational limits:** the DOT data has no gates, runways, fuel or crew. Use the new workbook (synthetic values; cited real figures also accepted).
- **New:** challenge2_operational_constraints.xlsx: per-airport gates, runways, movements per hour, turnaround; wake separation; fuel and crew rules.

Airbus Challenge #3: SAF supply chain in Kalimantan

- **Capacity:** biomass intake, for depots and refineries.
- **Yield:** 0.20 t SAF per t biomass. Depots lose no mass.
- **Demand:** 300 kt SAF per airport per year (1,200 kt total, needs 6,000 kt biomass). You do not have to use all 6,800 kt.
- **Costs:** one-year basis. CapEx straight-line over 20 years, no interest. Fixed OpEx 5% of CapEx per year; variable OpEx per tonne added to the facility table.
- **Distance:** great-circle × 1.3. Truck cost per tonne-km is the same for both sizes.
- **Airports:** Supadio (PNK), Iskandar = Pangkalan Bun (PKN), Sepinggan (BPN), Juwata (TRK). Tarakan is an island: last leg by barge at 0.08 USD per tonne-km.
- **New:** challenge3_parameters.xlsx and challenge3_airports_4.csv.

Airbus Challenge #4: Regulated technical information

- **Count:** 431 entries, not 432.
- **Type:** FAA proposed rules (NPRMs), not final ADs. Only 21 have a final AD number.
- **Revisions:** every row says "Current". Use the new revision_chains sheet: 99 dockets appear more than once; latest publish date = current version.
- **1,000+ pages:** summaries are ~150 pages. Fetching full documents from the Federal Register links is allowed and encouraged.
- **New:** challenge4_ad_index.xlsx: record IDs, clean dates, superseded ADs, serial-number flag, revision chains.

Airbus Challenge #5: Airport weather and delay risk

- **Wind direction:** was missing, so crosswind couldn't be computed. Now provided hourly for ATL and DFW, with headwind, crosswind and tailwind for headings 80° and 180°.
- **Main target:** delay over 45 minutes at ATL and DFW. Hourly AAR forecasting is now an **optional bonus** (the flights file is a sample).
- **No leakage:** only use weather observed at least 3 hours before the scheduled time you predict for.
- **Ignore:** `wpgt_*` and `snow_*` (placeholders).
- **Generality:** pipeline accepts any airport code; demo and score on ATL and DFW.
- **New:** challenge5_hourly_weather_ATL_DFW.csv and challenge5_data_guide.xlsx.

NUS Hackers

- **Submit:** public repo with an OSI-approved licence, a live demo under 3 minutes, and a README with sourced before/after costs.
- **Stay legal:** only targets you own or can lawfully access. No breaking DRM, bypassing paywalls or logins, or accessing others' accounts or data.
- **Clean-room:** don't copy proprietary code, assets, firmware or trademarks.
- **Privacy:** don't collect or publish others' data; strip your own keys before pushing.
- **Built today:** code written during the sprint (10:15 AM – 4:15 PM). Commit history is checked.
- **Evidence:** source prices and user counts; compare like for like, including running costs; licence must be compatible with your libraries.

Questions? Ask an organiser or post on Discord. Submissions close at 4:15 PM.